

# ACHINTYA RAI

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## EDUCATION

**University of California, San Diego**  
*Bachelor of Science in Computer Engineering*

Sept. 2024 – June 2028  
**GPA: 3.9/4.0**

## WORK EXPERIENCE

### Adobe

*Software Engineering Intern*

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks to establish the source-of-truth data pipeline.
- Designed a time-series anomaly detector, iterating over statistical algorithms to select a machine learning model handling multi-layer seasonality across **10+** workflows and **30+** countries.
- Built SQL and MLOps pipelines scheduling weekly retraining across **10+** machine learning models.

### Decklar (formerly Roambee)

*Software Engineering Intern*

Sunnyvale, CA

June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python and NumPy data pipelines to achieve a **15%** capacity boost across **1M+** weekly events.
- Implemented advanced data structures for analytics, yielding **20%** faster reporting speeds.
- Boosted workflow efficiency by **15%** via Panels-based UI/UX refinements and iterative testing.

### VALT Health

*Founding Software Engineer*

Remote

Aug. 2025 – Feb. 2026

- Engineered a Flask backend processing **100k+** datapoints from fitness devices, unifying data formats across **20+** distinct sources into normalized database schemas.
- Built asynchronous batch processing pipelines in GCP, yielding a **35%** gain in execution speed.
- Automated environment setup and data seeding via Bash, cutting testing time by up to **80%**.

## RESEARCH

### Graph Theory & Neural Architecture

*Undergraduate Research Fellow (NSF REU)*

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), implementing weight-matrix extraction to reduce **500GB** checkpoints to **13GB** localized manifolds.
- Optimized architectural placement scripts and preprocessing for a **30x** runtime reduction (30 mins to <1 min), enabling large-scale manifold analysis on resource-constrained **8GB RAM** systems.
- Applied **Rent's Rule** and Louvain Clustering to surface model backbones, raising Inter-Cluster Network density from **0.18%** to **13.25%** in Llama-2-70B via noise-filtering.

## PROJECTS

**MovieRecapinator** | *Java, Spring Boot, Maven, JavaScript, IMDb API* 

April 2025

- Developed a recommendation engine using **Spring Boot** to process **700,000+** entries from the IMDb API, identifying high-priority episodes for series recaps.
- Architected the full-stack system in **Java**, delivering a working MVP within a **24-hour** sprint.

**NanoGPT** | *Python, PyTorch, Transformers, Neural Networks* 

Mar. 2026

- Trained a character-level transformer chatbot with **100k+** parameters in **Python/PyTorch**, implementing multi-headed attention and backpropagation from scratch.
- Engineered scalable dataloaders to stream tokenized datasets at **100%** hardware utilization.

**SquaredUpChess** | *React, TypeScript, Django, WebSockets, AWS, Docker* 

Aug. 2025 – Dec. 2025

- Architected **20+ Django REST endpoints** for authentication, sessions, and transaction logic.
- Optimized **SQLite** via strategic indexing and query tuning for multi-user scalability.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Java, TypeScript, SQL (Postgres/SQLite), JavaScript, Bash

**Development:** React, Django, Node.js, FastAPI, Asyncio, NumPy, WebSockets, Flask, JUnit, PyTest

**AI & Data:** PyTorch, Machine Learning, Databricks, AWS, GCP, Docker, GitHub Actions, Linux